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We take more than a peek at the innards of the latest high-horsepower, pixel-grinding top guns as NVIDIA's G80 squares off against ATI's R600

Michael Browne

It's been an impressive sight to anyone even loosely following developments in the graphics industry. NVIDIA's G71 and ATI's R580 nearly doubled performance in all the games that were new back then, and visuals were improved dramatically. There was, however, a limit to what either vendor could do with DirectX 9.0's Shader Model 3.

Then Windows Vista was announced, along with the new, backwards-compatible DirectX 10. Speculation ran wild as everyone tried their hand at calculated guesses as to what the new standard would amount to in gaming terms. More importantly, who would emerge top dog from among NVIDIA and ATI—two companies that have been slugging it out since DX 8.1 days. With both companies now offering DX 10 capable GPUs, all speculation has been laid to rest. There is still uncertainty, because we still don't know how the latest cards perform under DX 10. Here, we hope to clarify things, however, by peeking under the bonnets of ATI's R600 and NVIDIA's G80.

A Demystification Of The Basics Shaders: Colouring at the speed of light!

As we all know, while rendering any 3D scene, texture (pixel) and geometry (vertex) information is sent to the graphics card for processing. Earlier, cards used to have a set of fixed algorithms hard-wired into them; this was used to process the data. It was known as the FFP (Fixed Function Pipeline). Game programmers could only select from the limited functions hard-wired onto those cards. Restrictions on the programmer and limitations in the flexibility of the code itself meant games couldn't look as realistic as developers wanted.

Shaders were developed in 2001, and their sole purpose was to give programmers more liberty with the code. There were primarily two types of shaders—*Pixel* and *Vertex*. On a software level, a shader is in essence a small algorithm that describes mathematically how a particular material or texture is rendered onto an object. This shader algorithm also specifies how light affects the appearance of the object. So a shader actually defines an object's physical properties in a 3D scene. This (shader) code is loaded into the graphics card's memory, and from there fed directly into the graphics pipeline, where it is executed.

